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Urban Pollution Governance, Prosecutor-led Environmental Public Interest Litigation and Regional Environmental Disparities in China: Evidence from 282 Cities

WU Wanqiang, Peter C.H. CHAN and LIN Xifen

Experts frequently discuss the reduction of regional environmental disparities in China, but they often overlook the role of the judiciary, particularly prosecutor-led environmental public interest litigation. Using cases of such litigation initiated by intermediate and basic procuratorates, and economic and pollution data of 282 cities for the 2018–21 period, this article demonstrates that prosecutor-led environmental public interest litigations have the potential to reduce China's environmental disparities in general. However, their implementation varies according to cities' respective level of economic development. This diversity arises from at least three primary factors: more affluent cities may possess more prosecutors with stronger environmental awareness; these cities have more litigation resources for this type of litigation; and they have greater procuratorial autonomy in the local economy, albeit weaker protection, that allows for initiating more prosecutor-led environmental public interest litigations. This article highlights the importance of prosecutor-led environmental public interest litigation in mitigating environmental disparities in China. It recommends increased investment in litigation resources, enhanced personnel training and greater judicial autonomy in less economically developed cities to amplify the impact of such litigation efforts in reducing such disparities.

Wu Wanqiang (aiden0517@sjtu.edu.cn) is a PhD candidate in Criminal Procedure Law at the KoGuan School of Law, Shanghai Jiao Tong University. His research centres on criminal justice and the procuratorial system.

Peter C.H. Chan (pchchan@cityu.edu.hk) is Associate Professor at the School of Law, City University of Hong Kong. He completed his PhD in Law at Maastricht University. His research focuses on law and society, empirical legal studies, Chinese law and comparative civil procedure.

Lin Xifen (linxifen1982@sjtu.edu.cn, corresponding author) is Professor and Associate Dean at the KoGuan School of Law and China Institute for Law and Society, Shanghai Jiao Tong University. He is also Professor at the China Institute for Socio-Legal Studies, Shanghai Jiao Tong University. He holds a PhD from the School of Law, Sichuan University. His research interests are criminal procedure, evidence law, prosecutorial reform, empirical legal studies in judicial process and decision-making, computational law, data-driven judicial artificial intelligence and urban governance.

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INTRODUCTION

China, as a low- to middle-income country,¹ has grappled with cross-regional disparities, a long-term critical issue that has affected millions of people. In response, the Chinese government has worked tirelessly to address such disparities. In October 2017, the Communist Party of China (CPC) revised the Constitution to include the view that “clear waters and green hills are mountains of gold and silver” (*lǚshuǐ qīngshān jiùshì jīnshān yīnshān*),² establishing it as one of the Party’s key governing ideas. Consequently, addressing environmental disparities among regions has become a top priority for all sectors in China.

As administrative oversight failures in environmental regulation become increasingly concerning,³ the government places great importance on legal instruments. Notably, the 2017 establishment of the prosecutor-led environmental public interest litigation (EPIL) system by the Supreme People’s Procuratorate (SPP) was a highly anticipated initiative among other varied judicial efforts. Immediately following the implementation, prosecutors across China initiated 1,737, 2,309 and 3,453 EPIL cases in 2018, 2019 and 2020, respectively.⁴ Prosecutorial authorities won over 99 per cent of these cases.⁵ The average annual increase of 41.42 per cent in litigation cases and an exceptionally high success rate indicate Chinese procuratorates’ active role in addressing environmental issues and achieving some success.⁶

Despite these developments, existing evidence may have not been sufficiently robust to convince all scholars of the effectiveness of China’s prosecutor-led EPIL in protecting the environment and reducing environmental inequalities between regions. For example, while some empirical studies have demonstrated the effectiveness of

¹ World Bank Group, “Data for Low & Middle Income, China”, at <<https://data.worldbank.org/?locations=XO-CN>> [13 April 2023].

² Gongchandang wang (www.12371.cn), “Zhongguo gongchandang zhangcheng (2017 xiuzhengan)” (Constitution of the Communist Party of China) (2017 Revision), 22 October 2022, at <<https://www.12371.cn/special/zggcdzc/?sw=dworb>> [31 January 2023].

³ Guo Wu, “On China’s Linkage Model of Environmental Administration and Environmental Judicature” (in Chinese), *Law Review* 2, no. 186 (2017): 183–96.

⁴ Supreme People’s Court of the People’s Republic of China (PRC), “Zhongguo huanjing ziyuan shenpan (2017–2018)” (China Environmental Resources Trials [2017–2018]), 18 October 2018, at <<http://wej.court.gov.cn/news/view-54.html>> [31 January 2023]; “Zhongguo huanjing ziyuan shenpan (2019)” (China Environmental Resources Trials [2019]), at <<http://wej.court.gov.cn/news/view-36.html#>> [31 January 2023]; “Zhongguo huanjing ziyuan shenpan (2020)” (China Environmental Resources Trials) (2020), at <<https://www.courtbook.com.cn/cxrm/33418.jhtml>> [31 January 2023].

⁵ Wu Wanqiang and Lin Xifen, “Constrained Power Expansion: China’s Procuratorial Reforms within and beyond Criminal Justice”, *Modern China* 5, no. 5 (2024): 285.

⁶ Lü Zhongmei and Zhang Zhongmin, “Huanjing sifa 2021: tuidong Zhongguo huanjing sifa tixijianshe maixiang xinzhengcheng” (Environmental Justice 2021: Advancing China’s Environmental Justice System on a New Journey), June 2022, at <<https://www.antpedia.com/attachments/att/file/20220628/1656383903909464.pdf>> [31 January 2023].

China's environmental judicial reforms in reducing pollution,⁷ sceptics contend that environmental protection has yet to become a core priority in China's political system and that these reforms' practical impact is further limited by weak local enforcement, bureaucratic overlap and local government interference.⁸ These institutional barriers are believed by such sceptics to significantly diminish the intended effects of China's environmental judicial reforms.

This article discusses the effects of prosecutor-led EPIL on environmental disparities by empirically examining prosecutor performance on such cases in cities with different economic and pollution levels. The authors compiled novel data on 7,142 prosecutor-led EPIL cases from 2018 to 2021, and economic development and pollution data from 282 prefecture-level and above cities between 2018 and 2021.⁹ Unlike past research focusing on single environmental pollution indicators,¹⁰ the authors employ a comprehensive and integrative set of comparable environmental pollution metrics. Methods include principal component analysis that synthesises the three variables of air, water and solid waste pollution into a single comparable variable, i.e. the environmental pollution index, as well as propensity score matching and linear regression analysis.

This article aims to address the following key questions: (i) Can prosecutor-led EPIL mitigate environmental inequalities in China? (ii) To what extent can prosecutor-led EPIL reduce these inequalities? (iii) What are the potential barriers to the efforts of prosecutor-led EPIL? (iv) How can procuratorial authorities overcome these obstacles to reduce environmental inequalities?

The article is structured as follows. The second section introduces the institutional background. The third discusses the data, variables construction and method. The fourth presents the findings and examines the impact of prosecutor-led EPIL on

⁷ For example, Chen, Shao and Wang's study of 143 cities showed that environmental public interest litigation significantly reduced industrial wastewater discharge. Chen Tianhao, Shao Jianshu and Wang Xuechun, "Evaluation and Improvement of the Prosecutorial Administrative Public Interest Litigation System: An Empirical Analysis Based on the Difference-in-Differences Method" (in Chinese), *Peking University Law Journal* 32, no. 5 (2020): 1328–52.

⁸ Xu Guangdong and Michael Faure, "Explaining the Failure of Environmental Law in China", *Columbia Journal of Asian Law* 29, no. 2 (2016): 1–96.

⁹ To the authors' knowledge, there are a total of 297 prefecture-level and above cities in China. However, after matching the caseload data with economic and pollution data, the authors obtained observation data for 282 cities over four years.

¹⁰ Zhang Junfeng, "Environmental Health in China: Progress towards Clean Air and Safe Water", *The Lancet* 375, no. 9720 (2010): 1110–9; Tu Peiyue, "Exposure and Inequality of PM_{2.5} Pollution to Chinese Population: A Case Study of 31 Provincial Capital Cities from 2000 to 2016", *International Journal of Environmental Research and Public Health* 19 (2022): 12137; Han Zhiyong et al., "A Review of Ground Water Contamination near Municipal Solid Waste Landfill Sites in China", *Science of the Total Environment* 569–570 (2016): 1255–64; Tang Jiexin, Wang Qunwei and Choi Gyunghun, "Efficiency Assessment of Industrial Solid Waste Generation and Treatment Processes with Carry-over in China", *Science of the Total Environment* 726 (2020): 138274.

China's environmental disparities and policy implications. The concluding section highlights the analysis and limitations.

INSTITUTIONAL BACKGROUND

This section examines the development, roles and challenges of EPIL in environmental protection and equity. It explores China's legislative initiatives to enhance environmental standards and discusses how local economic protectionism has hindered these efforts. It explains the establishment and benefits of prosecutor-led EPIL, emphasising judicial autonomy as a key solution to resolve local economic protectionism. The authors also establish a theoretical framework, following their review of the development issues and practices of EPIL, to argue that prosecutor-led EPIL is the most promising judicial approach under the current legal framework to addressing Chinese environmental issues.

Theory and Practice of Environmental Public Interest Litigation

Environmental public interest litigation serves as a crucial supplementary mechanism to protect the environment and promote environmental equity by mobilising citizens, social organisations and governmental bodies to take legal action against polluters in the public interest, after governmental regulations and market forces have failed to address environmental issues. Its inception can be traced to the environmental movements of the 1970s in the United States¹¹ when the necessity for legal action to protect public environmental interests became evident. The public directed their initial efforts towards addressing legislative and enforcement gaps that permitted environmental degradation to continue.¹² Over time, EPIL has evolved from its American origins into a global legal framework, and has also transformed from merely enforcing compliance with environmental laws to becoming a broader catalyst for environmental justice and social equity.¹³

A substantial body of related empirical research corroborates EPIL's efficacy in fostering environmental protection and equity. However, its strategic application could limit its efficacy. For example, in India, EPIL has proven effective in protecting the environment where special interest groups influence regulatory bodies or where corruption impedes enforcement. The Ganges River is one successful example of the

¹¹ Riley E. Dunlap and Angela G. Mertig, "The Evolution of the US Environmental Movement from 1970 to 1990: An Overview", *Society & Natural Resources* 4, no. 3 (1991): 209–18.

¹² Jacqueline Peel and Hari M. Osofsky, *Change Litigation: Regulatory Pathways to Cleaner Energy* (Cambridge University Press, 2015), p. 28.

¹³ Andrea Durbach et al., "Public Interest Litigation: Making the Case in Australia", *Alternative Law Journal* 38, no. 4 (2013): 219–23.

mitigation of air and water pollution.¹⁴ However, because EPIL often involves challenging powerful government entities or interest groups, resource-constrained social organisations, such as nongovernmental organisations (NGOs), may opt to focus on cases within specific and sometimes excessively high-risk domains.¹⁵

While engaging in high-risk cases can offer NGOs potential benefits like enhanced reputation and financial compensation, addressing a large number of “low-value” cases may exert a more substantial impact on environmental health. Furthermore, focusing primarily on “high-value” cases may inadvertently exacerbate environmental inequalities.

Although EPIL has achieved global success, its legal frameworks exhibit considerable variations between countries, particularly in theoretical doctrines and plaintiff qualifications. Certain aspects of these frameworks are not entirely conducive to the advancement of public-driven EPIL. For instance, in countries adhering to the Weberian civil law tradition, like Germany, the rule of law centres on the doctrine of individual public rights. This approach allows litigation only when administrative authorities violate a plaintiff's rights and thus lacks a system for judicial review based purely on the public interest.¹⁶

However, determining when a norm grants individual public rights is highly challenging and strict doctrine may restrict public participation in EPIL by mandating the government to protect the environment. Even in the United States, where greater public involvement is ostensibly encouraged by allowing anyone to initiate EPIL, numerous plaintiff qualification tests significantly restrict public involvement. For example, US courts often employ the ambiguous “zone of interests” test to restrict citizen engagement in EPIL cases.¹⁷

In summary, while efforts have been made in various legal jurisdictions to encourage public participation in EPIL, limitations posed by theoretical doctrines and existing or preceding laws suggest that advancing EPIL should not overlook the role of governmental departments and judicial organisations.

¹⁴ A.V. Raja and Francis Rathinam, “Economic Efficiency of Public Interest Litigations (PIL): Lessons from India”, Munich Personal RePEc Archive, Paper no. 3870, 2005, paper presented at the first annual conference of the Asian Law and Economics Association, Seoul National University, Seoul, South Korea; Michael G. Faure and Angara V. Raja, “Effectiveness of Environmental Public Interest Litigation in India: Determining the Key Variables”, *Fordham Environmental Law Review* 21, no. 2 (2010): 239–94.

¹⁵ Xie Lei and Xu Lu, “Environmental Public Interest Litigation in China: A Critical Examination”, *Transnational Environmental Law* 10, no. 3 (2021): 441–65.

¹⁶ Nicholas Campbell, “Individual Rights and the Environmental Public Interest: A Comparison of German and Chinese Approaches to Environmental Litigation”, *Review of European, Comparative & International Environmental Law* 32, no. 1 (2023): 105–18.

¹⁷ Bradford C. Mank, “Prudential Standing and the Dormant Commerce Clause: Why the Zone of Interests Tests Should Not Apply to Constitutional Cases”, *Arizona Law Review* 48 (2006): 23.

China's Environmental Legislation and Its Local Economic Protectionism

China initially aimed to address environmental issues by implementing increasingly stringent legislation and regulations at both national and local levels. The first environmental legal instrument, the “Environmental Protection Law (Trial)”, was enacted in 1979. Although this was a significant step forward in China’s environmental protection, the superficial legislative philosophy and techniques were inadequate to address the increasingly severe environmental challenges posed by economic growth and to meet international concerns about global environmental and development issues. The Constitution, revised in 1982, further emphasised environmental and natural resource protection, mandating the state to preserve and improve ecological and living environments. China’s legislative body amended the Environmental Protection Law to implement the Constitution and issued specific laws protecting the ocean, water resources and the atmosphere. However, the expanding legislation appeared ineffective in stopping the deterioration of China’s environment. In 2014, the Central Committee of the CPC proposed protecting the ecological environment with the strictest legal frameworks and again revised the Environmental Protection Law, along with approving and amending dozens of related laws. Beyond the central legislative efforts, local governments also actively formulated local regulations for environmental protection, averaging six environmental protection regulations annually between 1996 and 2004, with a notable peak in 2002 when 11 regulations were approved.¹⁸

Tightening environmental laws has prompted a positive response at the corporate level, especially among state-owned enterprises, which applied for significantly more environmental patents following the enactment of new environmental laws.¹⁹ However, despite intensified legislation and regulations, empirical studies suggest that the legislation may have limited effectiveness in improving environmental quality.²⁰ The enactment of local environmental regulations has not significantly reduced pollution emissions, possibly due to the enforcement reliance on local environmental bureaux whose capacity and dependence on local governments may constrain their effectiveness,²¹ as they tend to prioritise local economic development.

China’s local economic protectionism emerged after the “reform and opening up” period when China transitioned from a highly centralised planned economy to a market economy. With the central government’s decentralisation, local governments expanded their power, shifting from non-stakeholders to stakeholders in local economic development. This change led local governments to focus on central government

¹⁸ Bao Qun, Shao Min and Yang Dali, “Environmental Regulation, Local Legislation and Pollution Control in China”, *Environment and Development Economics* 26, no. 4 (2021): 321–39.

¹⁹ Liu Yuyu, Wang Aiguo and Wu Yuqing, “Environmental Regulation and Green Innovation: Evidence from China’s New Environmental Protection Law”, *Journal of Cleaner Production* 297 (2021): 126698.

²⁰ Bao, Shao and Yang, “Environmental Regulation, Local Legislation and Pollution Control in China”.

²¹ Ma Yi and Xiang Wen, “Enforcing Law Through Authoritarian Environmentalism? State and Non-State Actors in China’s Environmental Public Interest Litigation”, *Journal of Current Chinese Affairs* 52, no. 3 (2023): 464–87.

directives and boost the local economy. Moreover, the growth rate of the gross domestic product (GDP) and local fiscal revenue have long been key indicators in assessing subordinate officials by the superior government.²² Therefore, local officials employ every means available to ensure economic growth, including protecting high-value local industries from bankruptcy.²³

However, local economic protectionism poses potential risks to environmental protection and equity by causing wasteful duplication of construction resources and an excessive pursuit of economic aggregates, thus overlooking externalities of economic pursuit²⁴ like environmental damage. One effective solution to tackle economic protectionism is to decouple officials' career advancement from the economic growth of their regions.²⁵ Until the government is able to achieve this, investigating how an autonomous judiciary can curb local economic protectionism remains a crucial research area for legal economists. With environmental enforcement lagging behind environmental legislation, China may struggle to enhance environmental protection and equity through legislation and administrative regulation alone. Such a gap between enforcement and legislation could be a significant factor contributing to the rise of prosecutor-led EPILs.

China's Prosecutors and the Prosecutor-led Environmental Public Interest Litigation (EPIL)

Unlike their counterparts in the United States and Europe, who focus primarily on criminal cases,²⁶ Chinese prosecutors have the authority to handle cases outside of criminal matters. The Chinese Constitution positions procuratorates as neutral organs for legal supervision,²⁷ which broadly confers on them considerable powers and responsibilities. The Organic Law of the People's Procuratorate defines the procuratorate's duties, which include supervising police investigations, examining criminal cases and deciding on arrests, instituting public interest litigation and supervising litigation activities.²⁸ Essentially, a procuratorate's core function is to ensure the correct and uniform application of national laws (not just criminal laws) by exercising legal supervisory powers.

²² Zhang Weiying and Li Shuhe, "Interregional Competition and the Privatization of Chinese State-owned Enterprises", *Economic Research Journal* 12 (1998): 13–22.

²³ Ping Xinqiao, "The Incentives and Effects of Government Protection: An Empirical Analysis" (in Chinese), *Finance & Economics* 5 (2004): 3–10.

²⁴ Ibid.

²⁵ Alwyn Young, "The Razor's Edge: Distortions and Incremental Reform in the People's Republic of China", *The Quarterly Journal of Economics* 115, no. 4 (2000): 1091–135.

²⁶ Gwladys Gilliéron, *Public Prosecutors in the United States and Europe: A Comparative Analysis with Special Focus on Switzerland, France, and Germany* (Berlin: Springer, 2013), p. 1.

²⁷ "Zhonghua renmin gongheguo xianfa" (The Constitution of the People's Republic of China), 22 March 2018, at <http://www.gov.cn/guoqing/2018-03/22/content_5276318.htm> [31 January 2023].

²⁸ The National People's Congress of the PRC, "Zhonghua renmin gongheguo renmin jianchayuan zuzhifa" (Organic Law of the People's Procuratorate), at <http://www.npc.gov.cn/npc/c12435/201810/b59a94f891794d9980e950c5cd8a0204.shtml?utm_source=UfqNews> [31 January 2023].

For several decades after its founding, the PRC generally confined the legal supervision function of its procuratorial authorities to criminal justice, such as investigating corruption and misconduct, approving arrests and public prosecutions. However, in early 2019, to align with China's overall legal reform, the Supreme People's Procuratorate (SPP) introduced the "four major procuratorial activities" (*sida jiancha*) concept, classifying and expanding the procuratorial functions into criminal, civil, administrative and public interest litigation,²⁹ and gradually transitioning from an agenda focused on criminality itself to a balanced one that includes these four domains. Prosecutor-led EPIL is among the procuratorate's most vital and promising responsibilities beyond criminal litigation activities.

The "four major procuratorial activities" encourage Chinese prosecutors to be proactive and responsible in civil, administrative and public interest lawsuits, particularly public interest litigation. This commitment aims to form a stable exchange with the state, securing state support for expanding their powers.³⁰ Recognising the procuratorial interest, the Central Committee of the CPC issued the first document related to procuratorial work in June 2021, endorsing the procuratorates' participation in public interest areas.³¹ Consequently, besides being crime fighters and legal supervisors, prosecutors have become social governors, with environmental improvement as their key objective.³²

Once authorised to initiate public interest litigation, China's procuratorates emerged as pivotal forces in EPIL, despite their lower rank compared to other organisations within the legal framework. Under the Civil Procedure Law, prosecutors can file an EPIL case if ecological harm is detected and no other designated organisation can act.³³ The procuratorate should therefore remind other qualified organisations to initiate civil EPIL before directly filing an EPIL case. When dealing with environmental

²⁹ Lin Xifen and Wu Wanqiang, "Something Lost, Something Gained: Changes in China's Procuratorate in Response to the Reform of the National Supervision System", *China Law and Society Review* 6, no. 1 (2022): 79–110.

³⁰ The stable exchange with the state is crucial to the power expansion of the legal profession in China. See Liu Sida, *The Logic of Fragmentation: An Ecological Analysis of the Chinese Legal Services Market* (Nanjing: Yilin Press, 2017), p. 16.

³¹ The Supreme People's Procuratorate of the People's Republic of China, "Zhonggong zhongyang guanyu jiaqiang xinshidai jiancha jiguan falü jiandu gongzuo de yijian" (Opinions of the Central Committee of the Communist Party of China on Strengthening the Legal Supervision of Procuratorates in the New Era), 15 June 2021, at <https://www.spp.gov.cn/tt/202108/t20210802_525619.shtml> [31 January 2023].

³² Peter C.H. Chan and Wu Wanqiang, "From 'Line Appraisal' to 'Case-Process Ratio': Will the New Case Quality Assessment System Facilitate the Changing Role of Chinese Prosecutor?", *Hong Kong Law Journal* 54, no. 1 (2024): 212.

³³ Currently, these designated organisations are usually relevant authorities and qualified social public interest organisations specialising in environmental protection. The National People's Congress of the People's Republic of China, "Zhonghua renmin gongheguo minshi susongfa" (Civil Procedure Law of the People's Republic of China), at <<https://flk.npc.gov.cn/detail2.html?ZmY4MDgxODE3ZWQ3NjZlYTAxN2VINmFiOTlhZDFjYmM>> [31 January 2023].

pollution involving administrative agencies, the procuratorate should first advise the administrative authorities to enforce environmental laws. If these authorities fail to comply, the procuratorate can initiate an administrative EPIL.³⁴ Despite being the last in line, after the government and qualified social organisations, to file an EPIL case, the procuratorate is still highly effective. Prosecutors have initiated 95 per cent of EPIL cases and won nearly all of them, with amounts ranging from small sums to millions of dollars.³⁵ Currently, procuratorates are officially recognised by the Supreme People's Court (SPC) as the dominant force in EPIL.³⁶

The procuratorates' success in EPIL is attributed to three key factors. First, since social organisations face stringent legal restrictions on plaintiff qualifications and cannot initiate administrative EPIL,³⁷ their participation in such cases is limited. By contrast, procuratorates are not subject to these restrictions and can initiate civil and administrative EPIL nationwide. Second, EPIL cases often require years to conclude, involving extensive investigations, evaluations and legal fees,³⁸ posing significant challenges for many social organisations in China. Procuratorates, supported by tax revenue, do not face these financial constraints.

Third, the procuratorate, as a relatively powerful institution within China's legal system, is endowed with abundant litigation resources. While the authority to initiate EPIL has existed for less than a decade, prosecutors have swiftly developed an efficient litigation model, due to the high volume of cases and extensive litigation experience gained by practice.³⁹ Through the procuratorial system, they can also altruistically share this model with procuratorates in other regions. This established framework is far more effective than the approach typically used by social organisations, which often rely on temporary legal teams.

While the ultimate goal of EPIL is to maximise public participation, under China's current legal framework, procuratorates are the most effective means to enforce environmental laws and protect the environment, at least in the short term. They also

³⁴ The National People's Congress of the People's Republic of China, "Zhonghua renmin gongheguo xingzheng susongfa" (Administrative Procedure Law of the People's Republic of China), at <<http://www.npc.gov.cn/npc/c30834/201706/2579219236d64f59967f077e3a8cf739.shtml>> [31 January 2023].

³⁵ Xie Lei and Xu Lu, "Environmental Public Interest Litigation in China: Findings from 570 Court Cases Brought by NGOs, Public Prosecutors, and Local Government", *Journal of Environmental Law* 34, no. 1 (March 2022): 53–81.

³⁶ Lü and Zhang, "Huanjing sifa 2021: tuidong Zhongguo huanjing sifa tixijianshe maixiang xinzhengcheng" (Environmental Justice 2021: Advancing China's Environmental Justice System on a New Journey).

³⁷ To qualify, social organisations or environmental NGOs must be legally registered with the civil affairs departments of cities at or above the prefectural level. Furthermore, only those engaged in environmental protection public interest activities for over five consecutive years and who have not engaged in any illegal activities are eligible to file civil EPIL against actions that cause environmental pollution or ecological damage. See Article 58 of the Environmental Protection Act.

³⁸ Hao Zhuang and Steven A. Wolf, "Environmental Public Interest Litigation: New Roles for Civil Society Organisations in Environmental Governance in China", *Environmental Sociology* 7, no. 4 (2021): 393–406.

³⁹ Xie and Xu, "Environmental Public Interest Litigation in China".

hold potential to overcome local economic protectionism and promote environmental equality.⁴⁰ By focusing on cases that benefit the public interest rather than specific individuals or groups, prosecutors can reduce the impact of local politics on judicial decisions. Overall, addressing environmental issues and local protectionism in China remains a severe challenge that necessitates ongoing reform efforts, with procuratorate engagement being essential.

DATA AND METHODS

This article comprehensively investigates prosecutor-led EPIL in China’s intermediate and basic-level courts between 2018 and 2021. The authors attempt to compile all national prosecutor-led EPIL cases in these courts during this period. They collected 2018 to 2021 data on economic development and environmental pollution for 282 cities at prefecture level and above⁴¹ from the *China Statistical Yearbook* for those years. The authors chose to focus on cities at prefecture level and above because provincial-level panel data are limited in quantity and lack detail, while county-level panel data are overly abundant yet incomplete. Utilising city-level data is therefore the optimal choice.

Methodologically, we primarily employed two analytical approaches: principal component analysis and propensity score matching. Appendices I and II briefly outline the methodology used in this article.

Compilation of Caseloads of Prosecutor-led Environmental Public Interest Litigation (EPIL)

The data for this study on China’s EPIL cases initiated by intermediate and local procuracies were sourced online from *Fa Xin (Legal Information*, at <<https://www.faxin.cn>>). *Fa Xin*, established by the SPC and managed by the People’s Court Publishing Group, is China’s largest legal knowledge and case data platform. Its official authority and comprehensiveness were the primary reasons for choosing it for data collection.

The authors conducted a comprehensive retrieval of all public interest litigation cases initiated by basic-level and intermediate-level procuratorates in criminal, civil and administrative proceedings from 2018 to 2021. The specific search criteria are presented in Table 1. The resultant data set gathered for analysis includes prosecutor-led EPIL cases, totalling 1,259 in 2018, 2,270 in 2019, 3,436 in 2020 and 1,407 in 2021.

⁴⁰ Chen, Shao and Wang, “Evaluation and Improvement of the Prosecutorial Administrative Public Interest Litigation System” (in Chinese).

⁴¹ The administrative division of prefecture-level cities in China is a ranking slightly below the provinces and above the county level. These cities are not merely urban centres but also administrative jurisdictions that may include other cities, towns and rural areas within their boundaries. Additionally, China has four municipalities directly under the central government, which are at the provincial level and rank above prefectures, although they are formally considered large cities. This article includes them as well.

TABLE 1
DATA SEARCH CRITERIA

Type of Litigation	Data Search Criteria
Criminal Litigation	1. Adjudication years: 2018–2021. 2. Specific offences include: environmental damage crimes, major environmental pollution accidents, illegal disposal of imported solid waste, unauthorised import of solid waste, illegal fishing, illegal hunting and killing of precious and endangered wild animals, illegal acquisition, transportation and sale of precious and endangered wild animals, illegal hunting, illegal occupation of arable land, illegal mining, destructive mining, illegal logging, destruction of plants of national importance, theft of forest trees, indiscriminate logging, illegal acquisition and transportation of stolen and indiscriminately logged forest trees, illegal occupation of agricultural land, illegal acquisition, transportation, processing and sale of plants of national importance and products of plants of national importance, and environmental pollution. 3. Full text includes: “public interest litigation”. 4. Cases heard at: intermediate and basic-level courts.
Civil Litigation	1. Adjudication years: 2018–2021. 2. Specific causes of action include: liability disputes over pollution affecting the environment, air, water, noise, radiation, soil, e-waste, and solid waste. 3. Full text includes: “public interest litigation” and “procuratorate”. 4. Cases heard at: intermediate and basic-level courts.
Administrative Litigation	1. Adjudication years: 2018–2021. 2. Full text includes: “public interest litigation” and “procuratorate”. 3. Issues related to: the environment, atmosphere, water, noise, radioactivity, soil, e-waste, solid waste, light, and ecology. 4. Cases heard at: intermediate and basic-level courts.

Variables Construction

Environmental pollution is classified into three categories: air, water and solid waste. To measure air pollution, citywide emissions of sulphur dioxide (in tonnes) and nitrogen oxides (NO_x , in tonnes), as well as average annual concentrations of respirable particulate matter ($\mu\text{g}/\text{m}^3$) were used. These three variables were combined in principal component analysis (PCA) into a single air pollution variable. Water pollution was measured by the city’s industrial wastewater emissions (in million tonnes) and solid waste pollution was gauged by the city’s emissions of industrial solid waste that is not treated in an environmentally sustainable way (in million tonnes). These three pollution variables—air, water and solid waste—compose the single environmental pollution index (*Pollution*) using PCA. The logarithm of the city’s GDP was utilised to measure the level of local economic development. Descriptive statistics of the data are presented in Table 2.

The findings reveal that, on average, each locality had 6.54 environmental public interest litigation cases per year. About 13 per cent of city–year observations recorded no filing of EPIL cases in a given year and approximately five per cent reported more than 20 cases in a given year.

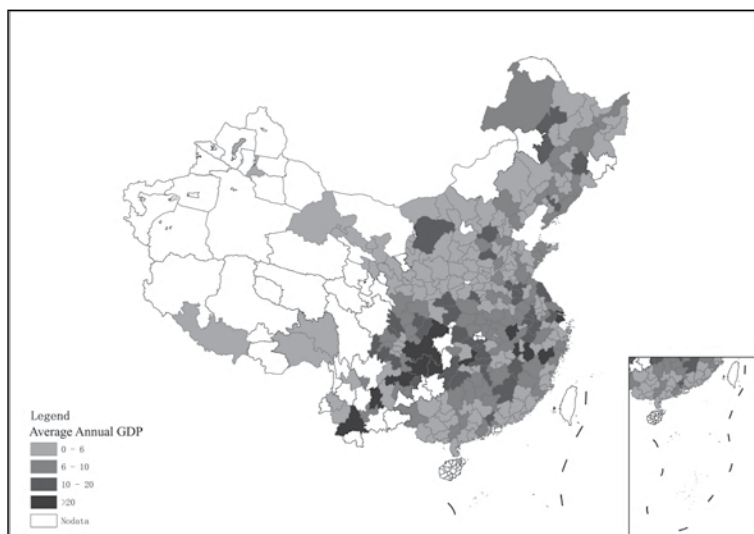
Figures 1 to 3 illustrate China’s average annual cases, pollution index and GDP by city between 2018 and 2021, with darker colours indicating higher values. The graphs show a strong correlation between the number of environmental public interest litigation cases, the level of environmental pollution and the degree of economic development.

TABLE 2
SUMMARY STATISTICS

	Mean	Std. Dev.	Min.	Max.
No. of cases	6.54	8.60	0	103
Pollution	7.39	1	3.57	11.5
Air Pollution	25.84	3.59	11.2	42.8
Water Pollution	2.07	0.68	0.79	9.65
Solid Pollution	232	67.2	104	1,859
Ln(GDP)	26.5	32.1	1.35	269.3

Note: Observations n=1,128.

Figure 1. Average Annual No. of Procuratorial Environmental Public Interest Litigation Cases by City, 2018–21



Source: Fa Xin (<https://www.faxin.cn>).

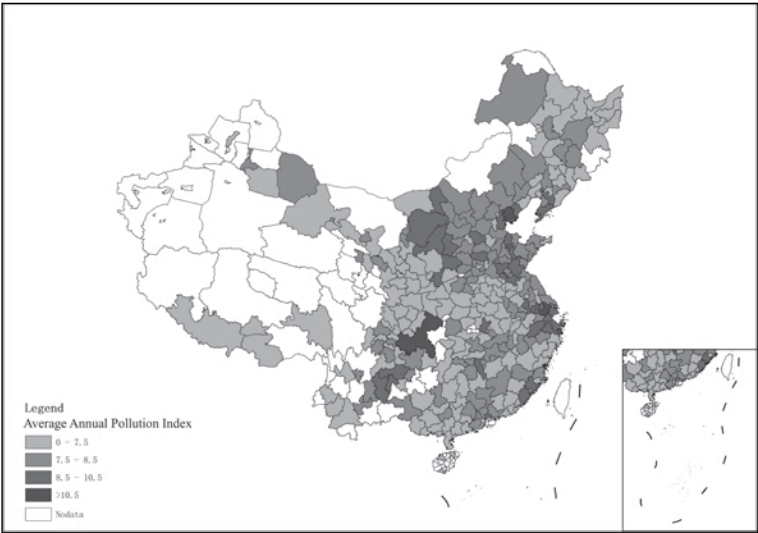
Results

Following the theoretical analysis of how environmental pollution levels impact local prosecutor-led environmental public interest litigation cases, the empirical effects were estimated using a fixed effects regression model, expressed as:

$$\text{Number of Cases}_{i,t} = \alpha + \beta \times \text{Pollution}_{i,t} + \text{City}_i + \text{Year}_t + \varepsilon_{i,t} \quad (1)$$

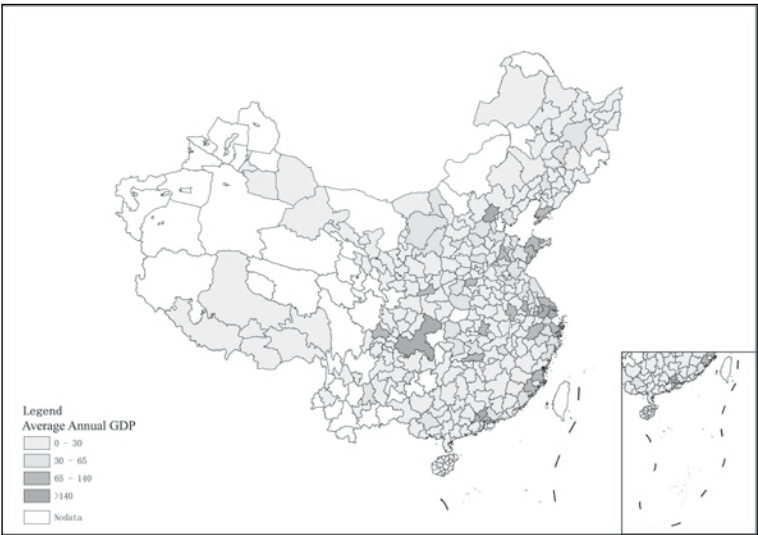
where i represents the city; t denotes the year; $\text{Number of Cases}_{i,t}$ denotes the number of environmental public interest litigation cases filed by the procuratorate in city i in year t ; $\text{Pollution}_{i,t}$ is the environmental pollution index of city i in year t ; $\text{Pollution}_{i,t}$ could also be replaced by $\text{Air Pollution}_{i,t}$, $\text{Water Pollution}_{i,t}$ or $\text{Solid Pollution}_{i,t}$ in different specifications. $\text{City}_{i,t}$ is the city fixed effect and $\text{Year}_{i,t}$ is the year fixed effect.

Figure 2. Average Annual Environmental Pollution Index by City, 2018–21



Source: *China Statistical Yearbooks* (<https://www.stats.gov.cn/sj/ndsjs/>).

Figure 3. Average Annual GDP by City, 2018–21



Source: *China Statistical Yearbooks* (<https://www.stats.gov.cn/sj/ndsjs/>).

The term α represents the intercept term while $\varepsilon_{i,t}$ represents the error term. The study particularly focuses on the coefficient of β . A positive β coefficient indicates that the case number of procuratorate-led EPIL increases with rising pollution levels, whereas a negative β coefficient suggests that procuratorate-led EPIL cases decrease despite heightened pollution levels.

The results of the model estimation are shown in Table 3. Results in column (1) clearly indicate that the level of urban pollution significantly and positively affects the number of local prosecutor-led EPIL cases, and suggests that as the level of urban pollution increases, the number of prosecutor-led EPIL cases also rises. The sub-analysis in columns (2)–(4) reveals that urban local air and water pollution levels also significantly and positively impact local prosecutor-led EPIL caseloads. Solid pollution, by contrast, has no significant effect on caseloads. These estimated results are consistent with the expectations in the preceding discussion: a locality's higher level of pollution correlates with a higher prosecutor-led EPIL caseload.

TABLE 3
IMPACT OF ENVIRONMENTAL POLLUTION ON THE CASELOAD OF ENVIRONMENTAL PUBLIC INTEREST LITIGATION

Dependent variable	(1) No. of cases	(2) No. of cases	(3) No. of cases	(4) No. of cases
Pollution	0.994*** (0.377)			
Air Pollution		0.355*** (0.094)		
Water Pollution			0.858*** (0.379)	
Solid Pollution				0.007 (0.009)
Obs	1,128	1,128	1,128	1,128
City fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Adj R ²	0.083	0.020	0.036	0.048

Notes: This table reports the impact of environmental pollution on the number of environmental litigation cases in cities. Air pollution includes emissions of sulphur dioxide (in tonnes), nitrogen oxides (NO_x, in tonnes) and respirable particulate matter (in µg/m³). Water pollution refers to the city's industrial wastewater emissions (in million tonnes). Solid pollution refers to the city's emissions of industrial solid waste that is not treated through an environmentally sustainable method (in million tonnes). "Pollution" is a composite measure of these three types of pollution. Robust standard errors are reported in parentheses. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

The impact of urban economic development on local pollution levels was estimated using the following formula:

$$Pollution_{i,t} = \alpha + \beta \times \ln(GDP)_{i,t} + City_i + Year_t + \varepsilon_{i,t} \quad (2)$$

where i represents the city and t , the year. $Pollution_{i,t}$ is the environmental pollution index of city i in year t . $Pollution_{i,t}$ could also be replaced by *Air Pollution* _{i,t} , *Water Pollution* _{i,t} or *Solid Pollution* _{i,t} in different specifications. $\ln(GDP)_{i,t}$ is the natural logarithm of the GDP of city i in year t . As in formula (1), $City_i$ is the city fixed effect and $Year_t$ is the year fixed effect. α represents the intercept term while $\varepsilon_{i,t}$ represents the error term.

Table 4 displays the estimation results of the model, indicating that the local economic development level significantly impacts environmental pollution levels, as reflected by the coefficients of β in all specifications of the pollution indicators. Specifically, higher local GDP correlates with an increased pollution level. From the

calculations for adjusted R^2 , it was found that economic development has the most significant effect on water pollution, with approximately 30 per cent of changes in water pollution being attributed to changes in economic development.

TABLE 4
IMPACT OF URBAN ECONOMIC DEVELOPMENT ON LOCAL POLLUTION LEVELS

Dependent variable	(1) Pollution	(2) Air Pollution	(3) Water Pollution	(4) Solid Pollution
Ln(GDP)	0.135*** (0.007)	0.337*** (0.030)	0.114*** (0.005)	1.363** (0.577)
Obs	1,128	1,128	1,128	1,128
City fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Adj R^2	0.189	0.089	0.304	0.004

Notes: This table illustrates the relationship between urban GDP development and environmental pollution levels. Air pollution is quantified by emissions of sulphur dioxide (in tonnes), nitrogen oxides (NO_x in tonnes) and respirable particulate matter ($\mu\text{g}/\text{m}^3$). Water pollution encompasses the city's industrial wastewater emissions (in million tonnes). Solid pollution includes the city's emissions of industrial solid waste that is not treated in an environmentally sound manner (in million tonnes). Overall pollution is represented as a combination of these three types. Robust standard errors are reported in parentheses. Significance levels are denoted by ***, ** and * for 1%, 5%, and 10%, respectively.

A propensity score matching (PSM) model was utilised to evaluate the impact of economic development on local environmental public interest litigation cases when pollution levels are similar. Cities were categorised into “high-GDP” and “low-GDP” groups depending on whether their GDP was above or below the median. High-GDP cities were then matched with low-GDP cities based on their air, water and solid pollution levels. Following the matching, the difference in the number of environmental public interest litigation cases between “high-GDP” and “low-GDP” cities was estimated.

Table 5 presents the results of the estimates, independent of propensity score matching. Given that both categories have similar pollution levels, the findings suggest that more economically developed cities tend to have a higher number of local environmental public interest litigation cases. On average, high-GDP cities had 3.21 times more local environmental public interest litigation cases than low-GDP cities did. This suggests that, given similar pollution levels, economically developed cities tend to focus more on environmental management and therefore have a higher number of prosecutor-led EPIL cases.

TABLE 5
IMPACT OF THE LEVEL OF ECONOMIC DEVELOPMENT ON THE NO. OF LOCAL ENVIRONMENTAL PUBLIC INTEREST LITIGATION CASES

Variable: No. of cases	High-GDP Cities	Low-GDP Cities	Difference
Unmatched	7.759	5.083	2.676*** (0.513)
Matched	7.728	4.515	3.213*** (0.931)

Notes: This table presents the difference in the number of environmental litigation cases between cities with high and low GDP after propensity score matching. ***, ** and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

DISCUSSION

Trend and Dynamics of Prosecutor-led Environmental Public Interest Litigation

Chinese prosecutors are inclined to pursue EPIL cases; however, the COVID-19 pandemic led to a significant decrease in these case numbers. The number of prosecutor-led EPIL cases from 2018 to 2021 were 1,259, 2,270, 3,436 and 1,407, respectively, reflecting growth rates of 80.3 per cent (in 2019), 51.3 per cent (in 2020) and a decline of 59 per cent (in 2021). It is reasonable to assume that the caseload number of prosecutor-led EPIL in China would have continued to grow if not for the stringent counter-COVID policies implemented by China's government in 2021.⁴²

In environmental protection, EPIL cases often serve as a last resort for China's procuratorates. Pre-litigation procedures, such as those classified as civil and administrative, typically resolve most environmental protection issues. For instance, civil pre-litigation proceedings constitute about twice the volume of litigation, while administrative pre-litigation proceedings are approximately 170 times the volume.⁴³ This ratio largely applies to EPIL as well.

With procuratorial resources remaining generally constant, there are three motivations for China's prosecutors to act in EPIL cases. First, from a national strategy perspective, EPIL has become an essential component of China's "Fourteenth Five-Year Plan for National Economic and Social Development (2021–2025)", and the procuratorate, as a state organ, is responsible for supporting this goal. Second, in terms of institutional interest, prosecutor-led EPIL helps fill the power vacuum left by the National Supervision Reform, which withdrew its original authority of investigating corruption cases.⁴⁴ Third, empirical evidence has shown that prosecutor-led EPIL effectively improves environmental conditions. This approach has strong support from China's government and official recommendation from the Central Committee of the CPC. Scholars familiar with China's political system understand that such recognition from the central authority could serve as an authoritative basis for expanding procuratorial powers in the future.

⁴² Zi Zhengfa and Wu Wanqiang, "Amendment of China's Biotechnology Laws in Relation to the Prevention and Containment of the COVID-19 Pandemic", *Biotechnology Law Report* 39, no. 6 (2020): 458.

⁴³ The Supreme People's Procuratorate of the PRC, "What Are the Highlights of Prosecution Work in 2018? Six Sets of Figures to Take You Through", 18 January 2019, at <<https://china.chinadaily.com.cn/a/201901/18/WS5c4198aaa31010568bdc4ce2.html>> [31 January 2023]; The Supreme People's Procuratorate of the PRC, "National Procuratorate Key Data Report for 2019", 2 June 2020, at <https://www.spp.gov.cn/spp/xwfbh/wsfbt/202006/t20200602_463796.shtml#1> [31 January 2023]; The Supreme People's Procuratorate of the PRC, "National Procuratorate Key Data Report for 2020", 8 March 2021, at <https://www.spp.gov.cn/spp/xwfbh/wsfbt/202103/t20210308_511343.shtml#1> [31 January 2023].

⁴⁴ Wu and Lin, "Constrained Power Expansion", p. 285.

Effectiveness of Prosecutor-led Environmental Public Interest Litigation in Reducing China's Environmental Disparities

Building on existing literature that underscores the potential of prosecutor-led EPIL in enhancing China's environmental protection, this study examines whether prosecutor-led EPIL can also reduce environmental inequalities. The study reveals two major findings. First, there is a direct correlation between economic development and environmental pollution. As shown in Table 3, analysis of 1,128 observations indicates a strong influence of local economic levels on environmental pollution, significant at the one per cent level ($p < 0.01$). Higher local GDP is associated with severely high pollution levels—this suggests that economically developed areas have a greater need for effective environmental governance.

Second, there is a positive correlation between pollution severity and the number of prosecutor-led EPIL cases. As is evident in Table 2, the number of EPIL cases positively correlates with overall pollution at the one per cent significance level ($p < 0.01$); this suggests that areas with more severe pollution tend to have more EPIL cases filed by prosecutors. The data also reveal that air and water pollution significantly and positively correlates with the number of prosecutor-led EPIL cases, while solid waste pollution shows no significant correlation. Such a discrepancy may be due to the more direct and perceptible impact of air and water pollution on people's daily lives and health, leading to greater public awareness and concern.

Overall, the statistically significant positive relationship between pollution levels and prosecutor-led EPIL cases indicates that prosecutors have channelled more resources and attention (through litigation) to areas with higher levels of pollution. Prosecutors generally demonstrate their responsiveness to the pollution control needs of their respective regions. In doing so, they are likely contributing to reducing environmental disparities between regions. While the regression results do not provide quantitative evidence of the extent to which prosecutor-led EPIL contributes to reducing environmental inequalities, the significant correlations suggest that prosecutors are dedicating more efforts to areas with more severe pollution; this indicates the general effectiveness of the prosecutor-led EPIL mechanism. If this system operates effectively and plays its intended role, it should, theoretically, help mitigate environmental disparities.

Deficiency of Prosecutor-led Environmental Public Interest Litigation in Reducing China's Environmental Disparities

Despite the potential benefits of prosecutor-led EPIL in mitigating environmental disparities in China, prosecutors' commitment to environmental protection varies significantly across cities. Among cities with comparable environmental pollution indices (Table 4), those that are economically developed tend to have a higher incidence of prosecutor-led EPIL cases. This suggests that these cities may benefit from better environmental protection, irrespective of the application of propensity score matching.

In view of the overall progress in environmental protection, there is a notable difference between poorer and affluent cities in the levels of environmental protection offered by procuratorates. Such a situation reveals that there may be a risk of widening environmental disparities between these cities.

One reason for this phenomenon could be that financially better-off cities can provide prosecutors with more resources, such as highly competent staff, professional investigative tools and sufficient funding to hire external assistance to file EPIL;⁴⁵ in short, prosecutors in more affluent cities have better resources to pursue environmental protection-related litigation. In addition, the leadership in procuratorial offices in economically developed regions may have a stronger understanding of environmental law and may be more supportive of initiating EPIL in cases involving environmental hazards. However, this impact is expected to wane with the implementation of the “provincial unified management reform”, as the provincial committee has uniformly allocated resources and leadership positions for procuratorates in local cities; thus, such a move would reduce the likelihood of resource and personnel imbalances.

Another explanation for the discrepancies in environmental protection is local economic protectionism. In economically underdeveloped cities, the local governments’ focus on economic development may undermine procuratorial authorities’ incentives to initiate EPIL against polluting enterprises that are vital to the local economy. This is particularly relevant as most of China’s grassroots cities have yet to develop and effect a transition to low-polluting high-tech or service industries, but still rely on highly polluting manufacturing industries as the backbone of their economies. This also explains the “high economy, high pollution” situation, as observed in Table 2. In such scenarios, officials in poorer cities, driven by economic achievement goals, may exert external influence on prosecutorial EPIL,⁴⁶ thereby reducing the volume of prosecutor-led EPIL cases in impoverished cities and widening their environmental disparities with wealthier towns.

In terms of policy implications, this article highlights three key aspects that require attention. First, at a macro level, maintaining the autonomy and authority of local procuratorial organs vis-à-vis local governments is essential. This approach would enable prosecutors to initiate and litigate environmental protection cases more effectively,

⁴⁵ Peter C.H. Chan, “Do the ‘Haves’ Come Out Ahead in Chinese Grassroots Courts? Rural Land Disputes between Married-Out Women and Village Collectives”, *Hastings Law Journal* 71, no. 1 (2019): 1–78.

⁴⁶ As Gao Qi has observed, local governments often intervene in local environmental cases due to economic interests. Several empirical studies support this view. For example, a study by Pang et al. found that local governments in China tend to adopt a “pollute first, control later” approach in balancing economic growth and environmental protection. Environmental regulatory measures become effective only when regional per capita GDP exceeds RMB90,000. Research by Wang and Wang reveals that the enforcement of emission fees is often inadequate in the most polluted areas. Hui et al. also demonstrate that cities under significant fiscal pressure are more likely to contribute to air pollution due to reliance on existing economic development models. These studies highlight two key points: (i) Environmental protection measures in poorer cities often have limited impact; and (ii) Local governments tend to

free from the undue influence of local officials. Second, regarding specific implementation programmes, in addition to continuing the “provincial unified management reform” and linking the career promotion of officials to environmental improvement, there is a need for local chief officials to move away from relying on high-pollution manufacturing as a main driver of economic growth. Recent empirical studies have shown that judicial reform, particularly enhancing the judiciary’s autonomy from local government influence, can also boost the local economy.⁴⁷ A fair justice system also promotes the corporate fixed assets investment rate, corporate financing, long-term investment, research and development (R&D) expenditures, and financial development.⁴⁸ By deepening judicial reform, prosecutor-led EPIL can more effectively reduce environmental inequalities in China. Third, at a micro level, it is vital to equip procuratorates in economically underdeveloped areas with the necessary resources and training to initiate and prosecute EPIL cases effectively. This approach would increase the volume of prosecutor-led EPIL cases in poorer cities, thereby narrowing the environmental protection gap between economically developed and underdeveloped regions.

CONCLUSION

Environmental disparities are critical issues faced by a country in its comprehensive development and long-term success, and they have become an increasingly important concern among environmentalists and jurists. Legal scholars have focused on whether and how the judicial system could contribute to environmental protection, but the issue of reducing disparities has been understudied. This article addresses this research gap by analysing prosecutor-led EPIL cases across all intermediate and grassroots procuracies, and also examining economic and pollution data from 282 cities between 2018 and 2021. It highlights the importance of reducing environmental disparities alongside enhancing overall environmental protection. By revealing a strong positive

prioritise economic development over environmental protection when under fiscal pressure. Consequently, since environmental public interest litigation is a fundamental measure of environmental protection, it may face similar challenges in achieving its goal in impoverished areas that are under the pressure of pursuing economic growth. See Gao Qi, “Public Interest Litigation in China: Panacea or Placebo for Environmental Protection?”, *China: An International Journal* 16, no. 4 (2018): 47–75; Wang Xianyan and Wang Lihong, “State-Enterprise Relation, Local Economic Priority, and Corporate Environmental Responsibility”, *Applied Economics* 51, no. 10 (2019): 995–1009; Pang Rui et al., “Pollute First, Control Later? Exploring the Economic Threshold of Effective Environmental Regulation in China’s Context”, *Journal of Environmental Management* 248 (2019): 109275; Hui Changhong et al., “Fiscal Pressure and Air Pollution in Resource-Dependent Cities: Evidence from China”, *Frontiers in Environmental Science* 10 (2022): 908490.

⁴⁷ Li Yishuang, “Court Structure, Judicial Independence, and the Attraction of Foreign Investment: Evidence from Judicial Recentralisation in China”, 8 April 2022, at <<https://ssrn.com/abstract=3994808>> [28 December 2022].

⁴⁸ Zhao Renjie and Zhang Jiakai, “Local Judicial System Reform and Corporate Investment: Evidence from Unified Management of Local Courts Below the Province”, *China Economic Quarterly International* 2, no. 4 (2022): 290–303.

correlation between environmental pollution levels and the volume of EPIL cases filed by prosecutors, this article validates the effectiveness of China's prosecutor-led EPIL in addressing environmental issues, thus providing evidence of its benefits in reducing environmental disparities. However, the different prosecution frequencies between cities with varying economic profiles may limit the potential impact of prosecutor-led EPIL.

The analysis demonstrates a positive correlation between pollution severity and prosecutor-led EPIL caseloads, indicating that prosecutors focus more EPIL efforts on heavily polluted areas. This implies that the prosecutor-led EPIL system is operating effectively and could contribute to reducing China's environmental disparities. In addition, high-GDP cities tend to have higher volume of prosecutor-led EPIL cases than low-GDP cities do, even with similar pollution levels. This suggests that while advancing environmental protection, China's procuracy has not succeeded in ensuring a balanced development between cities with different GDPs. This could be attributed to several factors: (i) cities with larger economic scale tend to have more prosecutors with stronger environmental awareness; (ii) affluent cities tend to have the capability to provide more resources, such as equipment support and professional assistance; and (iii) affluent regions often have greater judicial autonomy and less local economic protectionism, allowing prosecutor-led EPIL to be more effective. Nevertheless, in order to better address regional inequalities, environmental pollution levels should remain a primary consideration for procuratorial resource investment in environmental protection. In accordance with the economic theory of marginal utility, it can be argued that enhancing EPIL resource investment, personnel training and judicial autonomy in less economically developed cities could enable prosecutor-led EPIL to significantly reduce environmental disparities in China.

In addition to highlighting the potential of China's prosecutor-led EPIL in narrowing environmental disparities, this article also contributes to three literature streams. First, it adds to the ongoing discussion on China's environmental protection and disparities through a judicial approach, using an integrated environmental pollution index based on principal component analysis to estimate local pollution levels. Second, it enriches the debate on the effectiveness of prosecutor-led EPIL by detailing how it positively impacts the reduction of environmental inequalities. Third, it augments discussion on broader political and judicial topics in China, such as local economic protectionism, judicial autonomy and the evolving role of China's prosecutors.

Despite these contributions, this article has three significant limitations. First, although the fixed effects regression model shows significance at the one per cent level, it does not provide additional causal evidence. Future research may delve more deeply into the impact of prosecutorial public interest litigation on environmental inequality using a difference-in-differences approach. Second, the relationship between economic development and environmental pollution is complex. Even though the fixed effects model controls for city and year, more unobserved variables remain uncontrolled, thus potentially influencing the regression outcomes. Third, EPIL is a last resort for Chinese prosecutors in addressing environmental protection cases, as pre-litigation procedures,

particularly for administrative EPIL cases, accomplish a considerable percentage of the work. Therefore, future studies should incorporate pre-litigation proceedings as complementary data to offer a more comprehensive view of Chinese prosecutors' efforts to promote environmental equality.

APPENDIX I

PRINCIPAL COMPONENT ANALYSIS

Since, in multivariate statistics, an excessive selection of variables may lead to redundant or intersecting information due to the inevitable correlation between some variables, it is desirable to transform these indicators and represent them by a small number of new, unrelated indicators, i.e. potential conceptual indicators. The new variables enable researchers to obtain an overall indicator for ranking and classification purposes. Principal component analysis (PCA) uses dimensionality reduction to transform multiple indicators into one or a few composite indicators for the best synthesis and simplification of multivariate data while ensuring minimal loss of information.⁴⁹ Because pollution takes many forms and therefore can be expressed by a variety of indicators which need to be grouped together for clarity of analysis, PCA is considered the optimal method for quantitatively assessing the pollution level of cities.

The crux of PCA lies in its ability to achieve three fundamental objectives within the statistical analysis process: (i) elimination of correlation between original variables; (ii) determination of weights necessary for comprehensive evaluation; and (iii) reduction in the number of dimensions of comprehensive evaluation indicators.

Compared to single-indicator evaluation methods, comprehensive evaluation methods and fuzzy evaluation methods, PCA boasts several advantages. First, it employs multiple indicators for comprehensive evaluation rather than relying solely on a single indicator for one-sided evaluation, ensuring the utmost comprehensiveness in reputation evaluation. Second, PCA transforms correlated original variables into independent components, allowing for a comprehensive evaluation based on principal components. Additionally, the mathematical model of PCA indicates that the weights of each principal component in the comprehensive evaluation function are not artificially determined but rather determined according to the weight of the variance of each principal component in the total variance, thus eliminating the unreasonableness of artificially determined weights found in some evaluation methods and facilitating comprehensive evaluation. Finally, selecting k principal components through PCA enables retention of more than 80 per cent of the information contained in the original variables, resulting in simplified calculation and comprehensive evaluation. Based on this analysis, PCA was selected to evaluate the pollution levels of cities.

APPENDIX II

PROPENSITY SCORE MATCHING

It is assumed that there are n cities and each city i has two possibilities, i.e. gross domestic product (GDP) above or below the median. These possibilities correspond to the difference in the number of environmental public interest litigation cases in the same city when GDP is above the median ($Y_i(1)$) and the number of environmental public interest litigation (EPIL) cases when GDP is below the median ($Y_i(0)$),

⁴⁹ Karl Pearson, "On Lines and Planes of Closest Fit to Systems of Points in Space", *Philosophical Magazine* 2, no 11 (1901): 559-572; Harold Hotelling, "Analysis of a Complex of Statistical Variables Into Principal Components," *Journal of Educational Psychology* 24, no. 6 (1933): 417-41 and 498-520.

respectively. Then the difference for the same city is denoted as δ_i , representing the difference between the number of EPIL cases in a city with good economic development and the number of EPIL cases in a city with poor economic development.

$$\delta_i = Y_i(1) - Y_i(0)$$

The focus is more on understanding the differences in the city as a whole than the differences in individuals, as the analysis generally pertains to groups. So

$$ATT = E(\delta|D = 1) = E[Y(1)|D = 1] - E[Y(0)|D = 1]$$

where the variable D indicates whether economic development is better than the median.

$E[Y(0)|D = 1]$ represents the observation that if a city with a higher degree of economic development than the median was to have a lower degree of economic development than the median, this is a counterfactual and unobservable value. However, this can be calculated by fitting a dummy control group with propensity score matching.

The concept of the propensity score is introduced here, i.e. the probability that a city is more economically developed than the median, as follows:

$$P(X) = P(D = 1|X)$$

Here, X denotes a set of observable covariates.

In the case of using X to estimate the probability of urban development above the median, the ATT is estimated as follows.

$$ATT_{PSM} = E[Y(1)|D = 1, P(X)] - E[Y(0)|D = 0, P(X)]$$

The ATT_{PSM} here is the average treatment effect on the treated estimated using PSM.